

Rover deployment system for lunar landing mission



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ABSTRACT

For lunar surface exploration, a deployment system is necessary to allow a rover to leave the lander. The system should be as lightweight as possible and stored retracted when launched. In this paper, two types of retractable deployment systems for lunar landing missions, telescopic- and fold-type ramps, are discussed. In the telescopic-type system, a ramp is stored with the sections overlapping and slides out during deployment. In the fold-type system, it is stored folded and unfolds for the deployment. For the development of these ramps, a design concept study and structural analysis were conducted first. Subsequently, ramp deployment and rover release tests were performed using the developed ramp prototypes. Through these tests, the validity of their design concepts and functions have been confirmed. In the rover release test, it was observed that the developed lightweight ramp was sufficiently strong for a 50-kg rover to descend. This result suggests that this ramp system is suitable for the deployment of a 300-kg-class rover on the Moon, where the gravity is about one-sixth that on Earth. The lightweight and sturdy ramp developed in this study will contribute to both safe rover deployment and increase of lander/rover payload.

1. Introduction

Japan Aerospace Exploration Agency (JAXA) plans a lunar exploration mission that includes a lander and rover (See Fig. 1) [1,2]. At the beginning of this mission, a rover-carrying spacecraft (the lander) will land on the lunar surface and the rover will leave the lander for further exploration. A rover deployment system is essential for a lander to safely allow the rover to start on its journey.

Various behaviors of Mars/lunar landers have been widely studied. Algorithms of hazard detection/avoidance have been reported for landers in some studies (e.g., [3–5]). When a lander contacts with the Martian/lunar surface, the lander load should be safely supported by its landing gears/legs. To this end, contact dynamics between landing gear and ground were experimentally analyzed and modeled in Refs. [6,7]. Furthermore, crushable legs [8] and actively controllable legs [9] were proposed for a lander to absorb its landing shock. In many studies, it is assumed that a lander carries a rover on it; however, there are few studies on a deployment system to release the rover from the lander.

On Mars, exploration missions have been intensively conducted using various rovers. In these missions, the rovers landed on the Martian surface without being carried by landers. Mars Pathfinder Rover and Mars Exploration Rovers (MERs), which were both launched by Jet Propulsion Laboratory (JPL)/National Aeronautics and Space Administration

(NASA), landed on Mars using a parachute and airbag from the sky. After landing, a ramp that was almost horizontal to the ground was deployed, and the rovers descended this ramp onto the Martian surface [10,11]. For a large-sized rover (Mars Science Laboratory, MSL), a sky-crane system was utilized and the rover landed on the surface directly from the sky [12]. In these deployment methods, parachutes, which require an atmosphere to decelerate the landing speed, were utilized. Thus, the methods cannot be used in the same procedure on the Moon where there is a virtual vacuum.

Russia (under the former Soviet Union) and China have conducted lunar exploration missions using rovers carried by landers. In the early 1970s, Russia successfully deployed two rovers (Lunokhods 1 and 2) on the lunar surface [13–15]. For the rover deployment, two sets of ramps were deployed from a lander in two directions (e.g., left and right) and the rover descended one set of ramps. From the reports, although details of the ramp are not provided, it was observed that the rover attempted to drive down the steep ramp. In 2014, the Chinese rover (Yutu Rover) was deployed on the lunar surface using an elevator-type system [16]; however, the system has apparently not been reported in detail. In addition, a system using a crane was previously considered by our research group as the deployment method of a lightweight rover [17]. However, this system was deemed to be complicated and heavy especially for heavy-weight/large rovers. The deployment system should be as lightweight as

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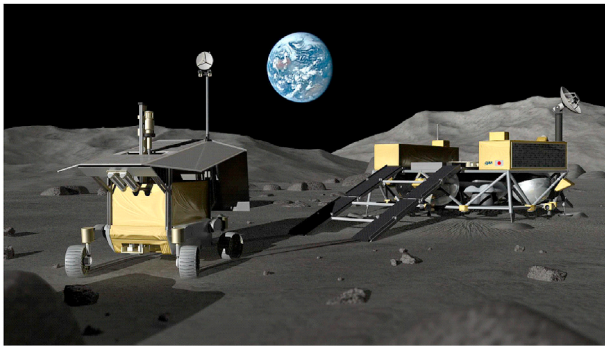


Fig. 1. Concept image of a lunar landing mission planned by JAXA that includes a lander and rover [1].

possible and stored retracted when launched.

Based on the above background, in this study, we discuss a ramp-type system for rover deployment. Two types of ramps, consisting of telescopic- and fold-type systems, were designed and their prototypes were developed. Furthermore, deployment and rover release tests were conducted using the developed prototypes. In Section 2, the requirements of a ramp for a lunar landing mission are discussed in detail. Subsequently, Sections 3 and 4 introduce the telescopic- and fold-type ramps, respectively. Section 5 compares the system characteristics of the telescopic- and fold-type ramps. Finally, Section 6 summarizes the conclusions.

2. Mission requirement for lander ramp

In general, lander ramps should be as light as possible and stored retracted when launched. Furthermore, the ramps must be reliable and ensure a redundancy, i.e., fault tolerance, to safely deploy a rover onto the lunar surface. A site contamination caused by the ramps and robustness of the ramp to the surface slopes/bumps should be also considered.

In this study, we designed a lander ramp that deploys a rover having a mass of approximately 300 kg. The load generated by this rover mass corresponds to approximately 50 kgf on the Moon, where the gravity is one-sixth that on Earth. It is necessary for the ramp to be sufficiently strong and as light as possible to support this load. In our current design of the lander, thrusters, which have a maximum height of 120 cm, are located on the lower part of the lander. Therefore, the ramp must be installed on the lander floor at this height above the ground. Considering rover traversability, the ramp inclination was determined as 20° . In this configuration, the ramp length was determined as $\left(\frac{120}{\sin 20^\circ} = \right)$ approximately 350 cm. Fig. 2 shows a concept image of the mission requirements for the target lander ramp system. As can be seen in Fig. 2, the area surrounded by lander footprints and extended ramps shall be free of hazards as the targeted area for landing.

When a lander is launched, the vibration generated can damage the ramp severely. To prevent this from occurring, a ramp having a length of 350 cm cannot be installed at its full length. In other words, the ramp should be stored retracted during launching and only deployed on the

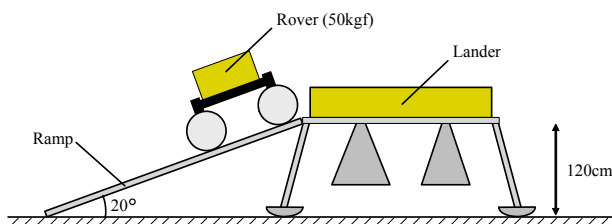


Fig. 2. Concept image of the mission requirements for the lander ramp system.

Moon surface. The length allowed for the stored ramp is determined based on a payload size/volume for a given mission. This length can differ depending on the design concept and mounting method of the ramp. In this study, we designed telescopic- and fold-type ramps having a length of 180 and 130 cm in their storage configurations, respectively.

3. Design and prototype development of telescopic-type ramp

We discuss two types of potential retractable ramp systems that fulfill the mission requirements: one is a telescopic-type ramp and the other is a fold-type ramp. In the telescopic-type ramp, the ramp that the rover descends is stored with the sections overlapping each other and slides out during deployment. In the fold-type ramp, the ramp is stored folded and unfolds during deployment. In this section, the telescopic-type ramp is described in detail along with its design concept, prototype development, and deployment test.

3.1. Design concept

As shown in Fig. 3, in the telescopic-type ramp, U-shaped plates are stored overlapping each other that slide out along slider rails attached to the side walls. In the storage mode shown in Fig. 3(a), it is assumed that the ramp is installed on the bottom floor of the lander. Thus, the length allowed for the stored ramp is determined based on the floor area. During deployment, two plates slide out in order as follows. First, a plate with the other plate inside slides out and becomes fixed to the lander floor by using hooks attached to its end. Bars are placed on each side of the lander floor and the hooks get fixed with the bar when the plate extends. Next, the second plate slides out and becomes fixed to the first plate by using latches and notches. The second plates have latches on their surface and the latches are locked with the notches on the first plate surface. Finally, the ramp is inclined around the bar under its own weight (See Fig. 3(b), (c), and (d)).

When a rover is ready to leave the lander, the ramp is deployed in a safe direction (e.g., front/back or left/right) based on the immediate topography around the landing point. For this reason, two sets of the same ramps needed to be mounted on the front and rear ends of the Russian lander in the past [13]. As shown in Fig. 4, one set of proposed ramps can be deployed in two directions along the slider rails. We believe that this concept of the ramp would contribute to reducing its total mass.

3.2. Prototype development and deployment test

We developed a prototype of the telescopic-type ramp based on its design concept and tested its designed function. In storage and deployment configurations, the ramp size were $180 \times 60 \text{ cm}^2$ and $350 \times 60 \text{ cm}^2$, respectively. A ramp having a length of 180 cm can be stored under the lander floor. For the function test in this phase, it is not necessary to match the material of the ramp with that of the flight model. Thus, the ramp was developed by using U-shaped plates made of aluminum with a thickness of 2 mm. The measured value of the ramp mass was 50 kg. While the ramp length was determined based on the mission requirement described in Section 2, its width and thickness differ from those of the final model.

Fig. 5 shows photographs of a deployment test using the developed ramp. In the photographs, the ramp was suspended at a height of 120 cm using a crane and wire. From the storage configuration, the plates first slid out and became fixed using a bar and hooks (See Fig. 5(a) and (b)). Next, the second plate was deployed and became fixed to the first plate using latches and notches. Finally, the ramp was inclined at an angle of 20° around the bar under its own weight (See Fig. 5(c)).

Furthermore, Fig. 6 shows photographs of the ramp deployed in two directions alternately. From the photographs, it can be observed that one set of ramps can be deployed in two directions using the same principle. Note that two pairs of the bar and hooks were equipped on both sides of the plates as shown in Fig. 3(a). The different pairs of them were used for

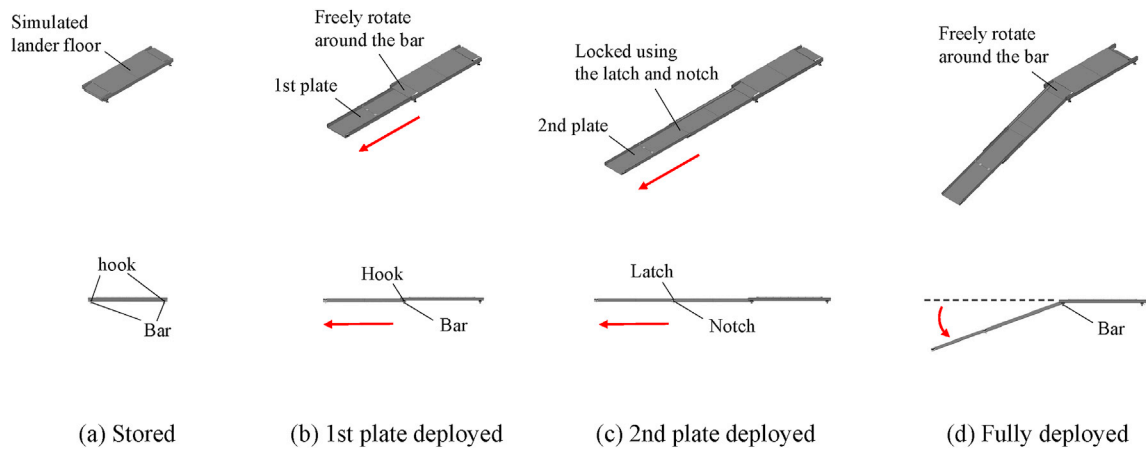


Fig. 3. Design concept of the telescopic-type ramp 1: overlapping U-shaped plates slide out for the deployment.

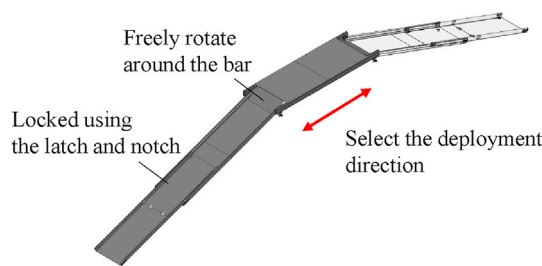


Fig. 4. Design concept of the telescopic-type ramp 2: one set of ramps can be deployed in two directions.

the deployment in two directions.

From the above description, the design concept of the telescopic-type

ramp was confirmed. In the test, an operator manually deployed the ramp, as shown in Fig. 5(b). In actual missions, we envisage that the ramp will be locked and released by using a spring/wire and separation bolts. If the separation bolts are installed on the both sides of the ramp and either one of the bolts is released, the ramp will be deployed in the direction selected. The spring/wire is used to control the deployment speed/sequence. We consider a separation mechanism similar to those presented in Ref. [18] as a potential system. The mass of these deployment mechanism is estimated as approximately 4 kg.

In the prototype developed, various mechanical components such as slider rails, hooks, and latches were made of aluminum/stainless and weighed approximately 10 kg in total. They can be used in their current shape/material in the actual mission. In other words, there will be no decrease in the mass of these components. Furthermore, although the dimensions of the ramp have not been finalized, on the basis of the CAD drawing, the mass of the current U-shaped plates were estimated at 6 kg

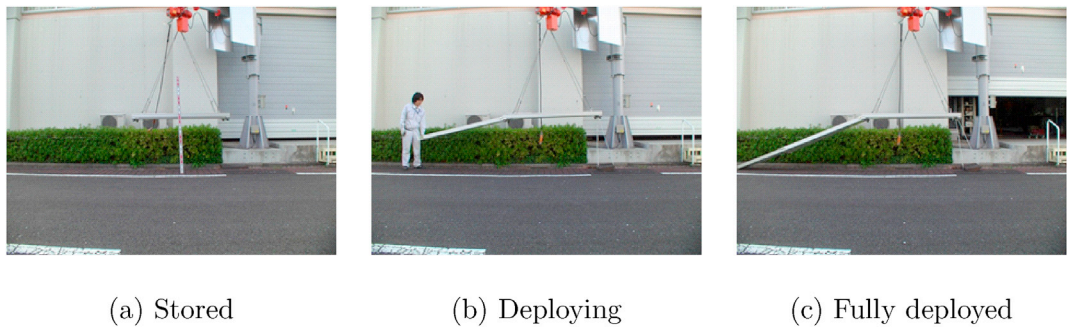


Fig. 5. Deployment test of the telescopic-type ramp 1: the ramp was suspended at a height of 120 cm using a crane and wire. In the test, the ramp was manually deployed by an operator.

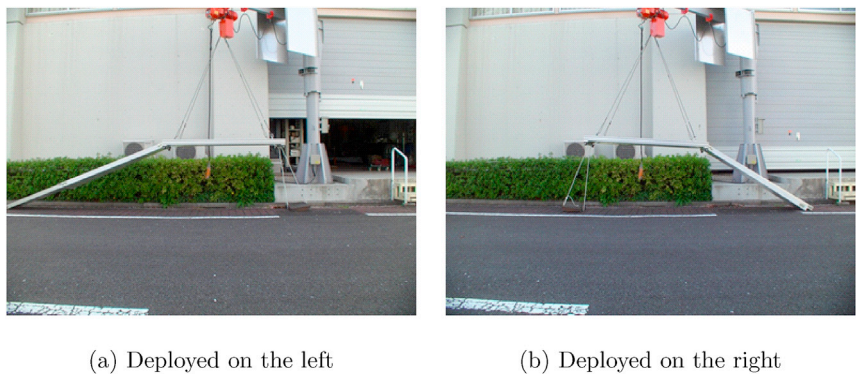


Fig. 6. Deployment test of the telescopic-type ramp 2: the ramp prototype was deployed in two directions.

using a carbon reinforced plastics (CFRP). We believe that no drastic increase in the plate mass from this estimation will be required. Based on these estimations, it is suggested that one set of the telescopic-type ramps could be developed with a mass of approximately 20 kg.

4. Design and prototype development of fold-type ramp

We designed and developed a prototype of the fold-type ramp. In this section, we describe in detail the fold-type ramp along with its design concept, structural analysis, and a rover release test using the prototype.

4.1. Design concept

In the fold-type ramp, L-shaped plates (i.e., rover driving plates with side walls) are stored folded together using hinges, as shown in Fig. 7. During deployment, the plates are unfolded at the hinges from the inside. For an ease of mounting, the ramp was designed with two folds. The plate nearest the lander body unfolds first and the lower plate unfolds second. The plates are fixed on base frames, and the hinges are attached to the frames. When the deployment is complete, the frame edge surfaces make contact with each other. This strengthens the L-shaped plate for a load/force from above. The plates must be deployed from the inside to obtain this benefit.

This ramp system can be deployed in only one direction. If there are any obstacles in the deployment direction, a rover cannot descend to the ground. Therefore, although the total mass of the system increases, two sets of the ramps must be mounted on the lander's front and back (or left and right) to prevent this from occurring. For this reason, finding ways of reducing the ramp mass is important especially for the fold-type ramp.

4.2. Structural analysis of ramp base frame

4.2.1. Structural analysis method

The L-shaped plates are required on the ramp to allow the rover to drive down and prevent it from falling off the ramp. However, if the rover mass can be supported by the ramp base frames to which the L-shaped plates are attached, the plates are not required to be excessively thick/strong, thus allowing the ramp system to be lightweight. That is, the lightweight structure of the ramp base frame is key to the mass reduction. Thus, we analyzed the deflection of the base frame and evaluated the influence of its deflection on the rover descent.

As shown in Fig. 7(c), when the ramp is deployed, assuming that the ramp system has a lock mechanism of its frames, its top end is fixed to the lander and its bottom end is supported on the ground. Based on this, we modeled the ramp frame as a beam fixed at one end and supported at the

other, as shown in Fig. 8. In the figure, the x-axis is parallel to the frame before deflected and its origin is fixed at the fixed end. For the beam with a concentrated load, P , its deflection, $\delta(x, P)$, can be expressed using the distance from the fixed end, x , as,

$$\delta(x, P) = \begin{cases} -\frac{1}{6EI} \left[R_0 \{ (L-x)^3 - 3L^2(L-x) + 2L^3 \} \right. \\ \quad \left. - P(L_2-x)^3 + PL_2^2(L_2-3x) \right] & (0 < x < L_2), \\ \frac{1}{6EI} \left\{ R_0 (3L^2(L-x) - (L-x)^3) \right. \\ \quad \left. - 3PL_2^2(L-x) \right\} & (L_2 < x < L). \end{cases} \quad (1)$$

Here, L denotes the beam length; while L_1 and L_2 denote the distances between the point of the concentrated load P and the supported/fixed end, respectively. R_0 denotes the reactive force required at the supported end, and is derived by

$$R_0 = \frac{PL_2^2}{2L^3} (2L + L_1). \quad (2)$$

Here, E and I denote Young's modulus and the second moment of area, respectively. For a frame having a hollow rectangular cross-section, I is expressed as,

$$I = \frac{1}{12} \{ BA^3 - (B-2t)(A-2t)^3 \}. \quad (3)$$

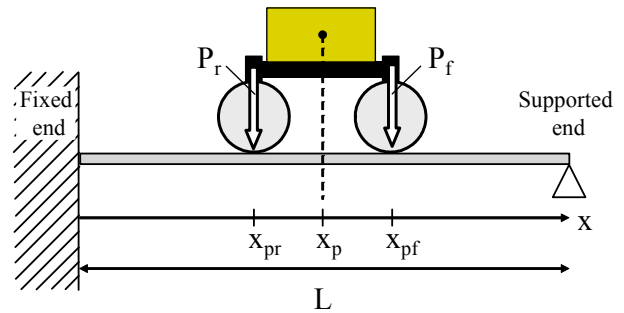


Fig. 8. Ramp frame model for the structural analysis. The frame was modeled as a beam fixed at one end and supported at the other.

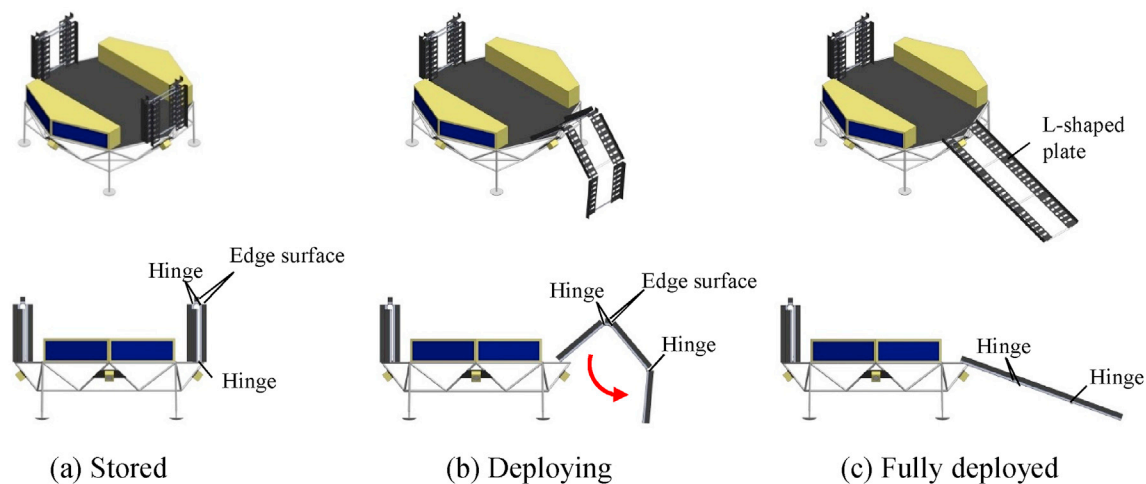


Fig. 7. Design concept of the fold-type ramp: L-shaped plates are unfolded from the inside.

Here, A , B , and t denote the height, width, and thickness of the hollow rectangle, respectively.

As shown in Fig. 8, when a rover drives down the ramp frame, its front and rear wheels make contact with the frame. Concentrated loads are developed at these contact points. Defining the distance between the fixed end and the center of the rover body as x_p , the frame deflection is derived by superpositioning the deflections generated by loads P_f and P_r that act on the points where the front and rear wheels make contact, as follows:

$$\delta(x_p) = \delta(x_p, P_f) + \delta(x_p, P_r). \quad (4)$$

Furthermore, defining the distance between the fixed end and front/rear wheel as x_{pf} and x_{pr} , respectively, the rover pitch, θ_{pit} , which corresponds to the angle of inclination of the rover from the horizontal line, can be derived as

$$\theta_{pit} = \sin^{-1} \frac{\delta(x_{pf}) - \delta(x_{pr})}{w_b}. \quad (5)$$

Here, w_b denotes the rover wheelbase, which can be calculated as $x_{pf} - x_{pr}$.

4.2.2. Structural analysis conditions

Using Equations (1)–(5), we evaluated the deflection of a frame having a hollow rectangular cross-section surface. In the evaluation, we assumed that a rover descended from the frame top end to the bottom end. Structural analysis was conducted with frames of different thicknesses and materials. The size of the cross-section area was fixed at $50 \times 20 \text{ mm}^2$, and the frame length was set at 350 cm. Aluminum and CFRP were used as the target materials. The analysis conditions are summarized in Table 1. In the table, CFRP A has the same mass as the aluminum by adjusting the thickness. Meanwhile, CFRP B has the same thickness as the aluminum.

In the structural analysis, the rover mass and size were set to correspond with those of the rover described in the following subsection. That is, the mass was set at 300 kg, which resulted in a force of 50 kgf on the Moon, where the gravity is about one-sixth that on Earth, and the wheelbase was set at 50 cm. We assumed that the load generated by the rover mass acted equally on the right and left ramp frames. Besides, on each frame, we assumed that the load was equally supported at the points where the front and rear wheels made contact with the frame.

4.2.3. Structural analysis results and discussion

Fig. 9 shows the structural analysis results obtained for ramp frames of different materials. These results show the frame deflection and rover pitch when the rover drives along the frame. From Fig. 9(a), it can be seen that the frame deflection (i.e., rover sinkage) increases as the rover travels along the frame and reaches a maximum at a distance of 2.05 m. Meanwhile, according to Fig. 9(b), the pitch angle reaches a maximum at a distance of 1.18 m.

From Fig. 9, it can be seen that the frame deflection and rover pitch were greatest for the aluminum. Although the maximum deflection of the aluminum frame was 33 mm, the change of the rover pitch was approximately $\pm 1.0^\circ$. This means that the rover pitch was a value between -21 and -19° over a frame with an inclination of 20° .

The data for the aluminum and CFRP A in Fig. 9 show that, by changing the material from aluminum to CFRP A, the frame stiffness increases without an increase in the mass. Meanwhile, the data for the aluminum and CFRP B show that, by changing the material to CFRP B, the frame mass can be reduced to approximately 55% of that of the aluminum frame, and that this also contributes to a decrease in the rover pitch.

According to the above results, when a rover having a mass of 300 kg descends a ramp frame with a length of 350 cm on the Moon, the designed frame will have sufficient stiffness and the change of the rover pitch will be within $\pm 1.0^\circ$.

4.3. Development of ramp frame prototype

Based on the above structural analysis, we developed a ramp frame prototype. This frame was made of aluminum angle pipe. The frame cross-section area was a hollow rectangle having an area of $50 \times 20 \text{ mm}^2$ and a thickness of 1 mm. Figs. 10 and 11 show schematic illustrations and photographs of the developed ramp frame, respectively. The ramp sizes were $130 \times 90 \text{ cm}^2$ and $350 \times 90 \text{ cm}^2$ for the storage and deployment configurations, respectively. The measured value of the mass was 5.8 kg. This ramp was developed based on the same design concept as that shown in Fig. 7, except that it did not have the L-shaped plates.

In the rover release tests described below, the ramp was manually deployed. In actual missions, the ramp will be locked with separation bolts in the storage configuration, and unfolded by releasing these bolts with springs/wires. Moreover, as an alternative method of deployment, we envisage a system using wires controlled with actuators.

4.4. Rover release test

4.4.1. Test overview and conditions

Rover release tests were conducted using the ramp prototype to verify the strength and stiffness of the ramp frame. Fig. 12 shows the rover used in the tests. This rover was equipped with four wheels and drive/steer motors were mounted with Harmonic Drives on each wheel. It was driven and steered using motor encoders and drivers (ZDCMD-S/CE $\times 8$; ZUCO). A gyroscope (CRH02-25; Silicon Sensing) and accelerometer (CXL17LF3; Sumitomo Precision Products) were mounted on the center of the rover body to measure the attitude. The power was supplied from the mounted batteries (DUO-150; IDX) and data was sent/received in a serial communication from a laptop. The rover specifications are listed in Table 2.

In the tests, as shown in Fig. 12, the ramp frame having a length of 350 cm was fixed to a platform with a height of 120 cm. In this configuration, the ramp inclination was set to 20° . The rover descended the ramp frame from its top end to the bottom end. As listed in Table 2, the rover mass was 30 kg. A 20 kg weight was added to the rover to simulate the behavior of a 300-kg rover on the Moon (i.e., a 50-kgf rover). The rover traveling speed was set at 1 cm/s and the steering angle was controlled at 0° . Using the gyroscope and accelerometer mounted on the rover, the rover attitude was measured in a cycle of 1 Hz.

In the tests, the rover was suspended from a crane using wires to prevent it from falling from the ramp. By manually adjusting the lengths of the wires and keeping them loose, the rover mass was always completely supported by the ramp (i.e., the wires did not lift the rover).

Table 1

Parameters of various materials used in the structural analysis. The mass was calculated for one frame having a length of 350 cm and hollow rectangular cross-section area of $50 \times 20 \text{ mm}^2$.

	Symbol	Aluminum(A5052)	CFRP A	CFRP B	Unit
Young's modulus	E	7200	12,000–15,000	12,000–15,000	kgf/mm ²
Density	ρ	2.7	1.5	1.5	g/cm ³
Thickness	t	1.0	1.9	1.0	mm
Mass	m	1.3	1.3	0.7	kg

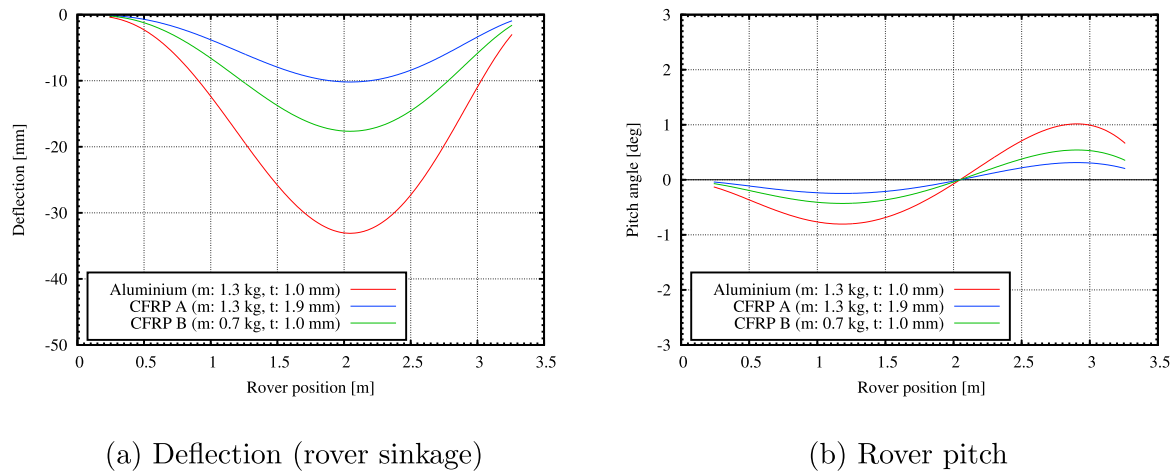


Fig. 9. Structural analysis results for the ramp frame. The horizontal axis represents the rover position on the ramp frame from the top end (i.e., fixed end).

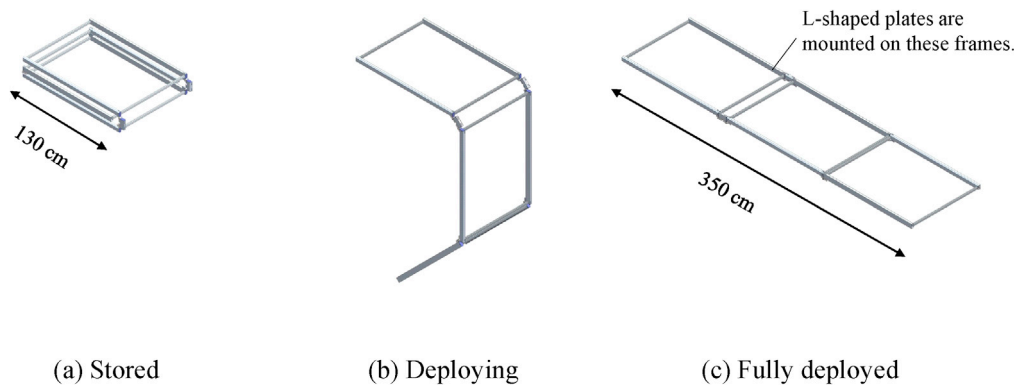


Fig. 10. Schematic illustrations of the frame for the fold-type ramp. This frame was developed based on the same design concept as that of the ramp shown in Fig. 7 except that it did not have L-shaped plates.

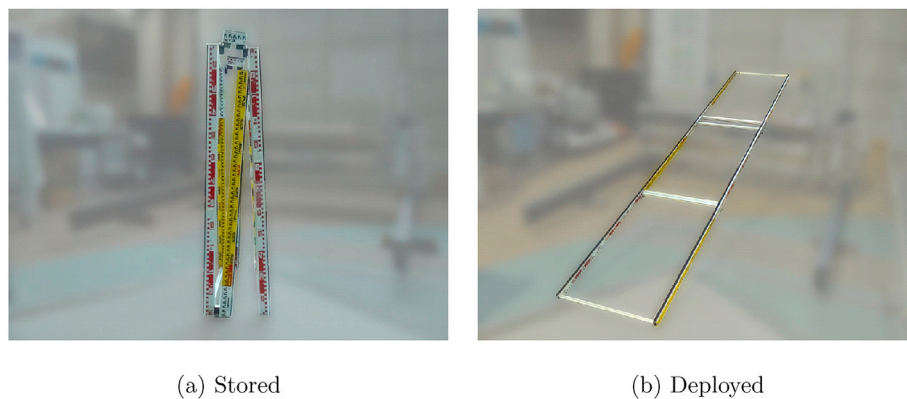


Fig. 11. Photographs of the frame for the fold-type ramp. In (b), the ramp top end was fixed on a platform with a height of 120 cm.

In preliminary tests, the rover wheels slipped and the rover almost fell from the ramp because of the low friction between the rover wheel and frame. Although in actual missions the surface material/geometry of the ramp can be changed to prevent this from occurring, in this test, a sticky sponge was simply installed on the frame surface. As the purpose of the tests was to verify the ramp strength and stiffness, these adjustments did not affect the verification results.

4.4.2. Test results and discussion

In this section, we discuss the ramp deflection from the rover attitude

data. Fig. 13 shows the change in the rover angle as the rover traveled along the frame. In the figure, the horizontal axis represents the rover position (i.e., traveling distance) from the top end of the frame. It can be seen that when the front wheel made contact with the ground at a traveling distance of 3.3 m, each value changed drastically. Subsequently, when the rear wheel made contact with the ground at a traveling distance of 3.8 m, the rover pitch became almost 0° , as expected.

From Fig. 13, it can be observed that the rover roll was almost constant as the rover traveled down the ramp (a value between -1.6 and 1.7° as listed in Table 3). Meanwhile, it can be seen that the pitch became

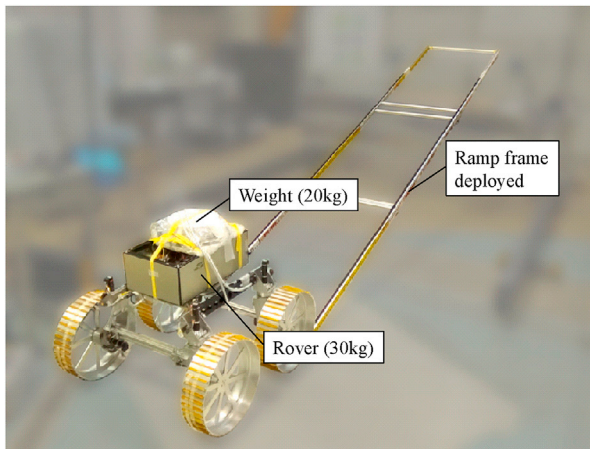


Fig. 12. Photograph of the rover release tests. In the tests, the rover was suspended from a crane using wires to prevent it from falling from the ramp. By manually adjusting the lengths of the wires and keeping them loose, the rover mass was always supported completely by the ramp.

Table 2

Rover specifications. In the rover release tests, the rover mass was set at 50 kg using additional weights.

Item	Value	Unit
Mass	30 (+20)	kg
Wheelbase	50	cm
Track width	65	cm
Wheel diameter	38	cm
Wheel width	16	cm

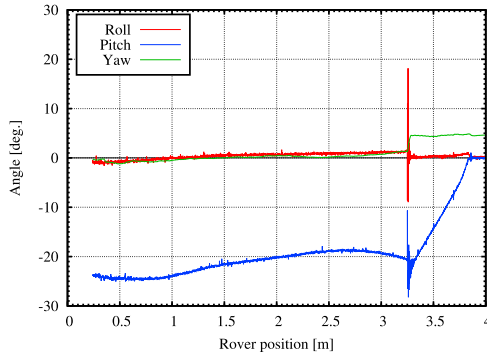


Fig. 13. Rover attitude angle vs. position on frame in the release test. The horizontal axis represents the rover position (i.e., traveling distance) on the ramp frame from its top end.

steeper as the rover traveled from the top end to the midpoint and subsequently became gentler. This trend corresponded with that of the structural analysis shown in Fig. 9(b). The value of the pitch was between -25.0 and -17.9° , as listed in Table 3. The yaw was almost constant throughout the test (a value between -1.2 and 1.4°). From the above description, it was confirmed that the frame was not severely deflected. In the rover release tests using the ramp frame prototype, the rover

pitch reached a maximum of 25° . This inclination was larger than that estimated from the structural analysis. In the structural analysis, the frame was assumed to be a long rigid beam without any joints. However, the developed prototype had joints between the frames. Hinges that consisted of joints were fixed to the frame using screws and nuts. When the frame was deformed, the screws may have slackened, resulting in further deformation of the frame. In future tests, the rover pitch could be reduced by improving the joints, e.g., integrated fabrication of the frame and hinges. In addition, the structural analysis shown in Fig. 9 suggests that changing the frame material would also contribute to a decrease in variation of the rover pitch.

In the development of the ramp frame for a 300-kg rover released on the Moon, we believe that no drastic increase in the frame mass from the above prototype will be required. That is, one set of the ramp frames will be developed with a mass of approximately 6 kg. In addition, on the basis of the CAD drawing shown in Fig. 7, the mass of the L-shaped plates was estimated at 5 kg. Based on these estimations, it is suggested that one set of the fold-type ramps could be developed with a mass of approximately 15 kg, including the deployment mechanism and separation bolts. For the fold-type ramp, two sets of the ramps will need to be mounted on the lander. Thus, the total mass of the ramp system is estimated at $30 (= 15 \times 2)$ kg.

5. Comparison of telescopic- and fold-type ramps

As explained in Section 2, we consider that a mass, ease of mounting, reliability, and redundancy (i.e., fault-tolerance) as important criteria for a rover deployment system. Table 4 provides a comparison of the telescopic- and fold-type ramps in these criteria. From the table, it can be observed that, although the fold-type ramp may be inferior to the telescopic-type in terms of mass and ease of mounting, it is superior in terms of degree of reliability and redundancy. In this section, the reasons of the rating in the table is explained in detail.

For the telescopic-type ramp, while its prototype weighted approximately 50 kg, the mass of the final system was estimated at 20 kg. This mass was estimated on the assumption that the dimension and material can change for the final system. On the other hand, for the fold-type ramp, while its ramp frame prototype weighted approximately 6 kg, the mass of the one set of final system was estimated at 15 kg. For two sets of the ramp, the total mass is 30 kg. The mass was increased in the final system because only the ramp frames were developed as the prototype and various additional components are essential in the final system. Based on the expected mass of the final systems, the telescopic-type ramp will be lighter than the fold-type ramp. For this reason, a mass was rated as “good” for the telescopic-type ramp in Table 4.

The telescopic-type ramp can be stored under the lander floor and one set of the ramps can be deployed in two directions. Meanwhile, two sets of the fold-type ramps need to be mounted on the front and rear ends of the lander. Thus, the total volume of the fold-type ramp is larger than that of the telescopic-type ramp. Furthermore, considering the storage configuration, it seems that the telescopic-type ramp can be more compact and have less impact on the lander total design than the fold-type ramp. For this reason, an ease of mounting was rated as “good” for the telescopic-type ramp in Table 4.

Note that regolith affects mechanisms of each ramp on the lunar surface. For the telescopic-type ramp, the regolith could become adhered

Table 3

Average, minimum, and maximum values of the rover attitude angles in the release test. Each value was obtained from the data before the front wheel of the rover made contact with the ground at the rover position of 3.3 m shown in Fig. 13.

	Roll [°]	Pitch [°]	Yaw [°]
Average	0.4	−21.4	0.03
Minimum	−1.6	−25.0	−1.2
Maximum	1.7	−17.9	1.4

Table 4

Comparison of the telescopic- and fold-type ramps.

	Telescopic-type	Fold-type
Mass	Good	Poor
Mounting	Good	Poor
Reliability	Poor	Good
Redundancy	Poor	Good

to the surface of the slider rails of the ramp. This could prevent the sliders from deploying smoothly. In addition, if the ramp system experiences a problem, the rover would not be able to leave the lander because the rover deployment depends on only one set of ramps. That is, this system has a low degree of redundancy. Owing to these characteristics of the telescopic-type ramp, its degree of reliability and redundancy were rated as “poor” in Table 4.

For a lunar landing mission, the lander systems must be developed as lightweight and compact as possible on the assumption that they have a high degree of reliability and redundancy. That is, in Table 4, the reliability and redundancy have higher priorities than the mass and mounting for the final decisions. Therefore, we envisage the fold-type ramp, rather than the telescopic-type ramp, to be a promising potential system for the mission. In addition, for the fold-type ramp, numerical simulations can be conducted using the proposed model for various conditions (e.g., the ramp size, rover size/mass and material). This will contribute to design a proper ramp for a given rover.

6. Conclusions

In this study, we discussed deployment systems that are necessary for a rover to be able to leave the lander in lunar landing missions. Telescopic- and fold-type ramps were introduced as potential systems. For the telescopic-type ramp, we developed a prototype and conducted deployment tests. The tests verified its design concept: (1) the U-shaped plates stored overlapping each other could slide out and be deployed, and (2) one set of ramps could be deployed in two directions. For the fold-type ramp, we conducted structural analysis, developed the prototype, and performed rover release tests. The release tests indicated that the developed ramp frame would allow a 300-kg rover to be deployed on the Moon. These tests suggested that one set of ramps could be developed with a mass of approximately 15 kg. The lightweight ramp developed will contribute to both safe rover deployment and increase of lander/rover payload.

In future studies, ramp deployment methods will be designed in detail based on scale model experiments and numerical simulations. Furthermore, the ramp system mass can further be reduced by optimizing its shape and material comprehensively.

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